

SafeHer: AI- Driven Predictive Model for Women Safety and Awareness

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Abstract—The issue of women’s personal safety during urban navigation remains significantly neglected by current mapping technologies. Current systems such as Google Maps and Apple Maps optimise routes exclusively for travel time and distance, providing no mechanism for crime-risk-aware routing. This paper presents SafeHer, a real-time safety navigation framework for women that addresses this gap through two principal contributions: (1) a Crime Danger Index (CDI) target formulation that resolves class imbalance in historical crime data by converting per-cell violent crime statistics into a uniformly distributed percentile rank, enabling robust regression; and (2) a two-component risk architecture that separates learnable spatial danger—modelled by a LightGBM regressor trained on 8.4 million Chicago Police Department incidents spanning 2001–2025—from volatile temporal modulation captured by a 168-slot empirical violent-crime-rate lookup covering all 24 hours across all 7 days. The combined model achieves $R^2 = 0.9997$ and $MAE = 0.003$ on held-out grid cells, while regenerating a full 14,129-cell city-wide heatmap in under one second—over 100 times faster than CNN-LSTM baselines. A unified risk grid architecture ensures that heatmap colours and route safety scores are always computed identically, eliminating the mathematical inconsistency present in all prior systems. A React-based frontend with Google Maps integration shows how temporal heatmaps change in real time and how to score safe routes.

Index Terms—women’s safety, crime prediction, LightGBM, temporal heatmap, safe navigation, spatial-temporal modelling

I. INTRODUCTION

Women living in cities are still very worried about their own safety. Empirical research consistently indicates that fear of crime substantially restricts women’s mobility, curtails their engagement in public life, and necessitates route alterations that prioritize perceived safety over efficiency [1]. Despite this documented necessity, prevalent navigation platforms—such as Google Maps, Apple Maps, and Waze—exclusively determine routes based on travel time or shortest distance, lacking any provision for safety-conscious route selection.

Existing safety-oriented applications address this gap only partially. Crowdsourced platforms such as Safetipin [2] rely on subjective user audits that are geographically sparse, temporally static, and dependent on sustained community participation. Academic systems including SafePath [3] and Crime-Travel [4] construct crime-weighted route graphs but suffer

from two structural limitations that SafeHer is specifically designed to overcome: (i) their risk models do not account for temporal variation, treating a zone as equally dangerous at noon and at midnight; and (ii) their map visualisation and route-scoring components use independently-computed risk estimates, producing mathematical inconsistency between what a user sees on the heatmap and the risk assigned to a route through the same zone.

SafeHer addresses both limitations through a unified risk architecture grounded in 24 years of officially reported crime data from the Chicago Police Department. The key insight is that heatmap visualisation and route scoring are fundamentally the same operation—given a geographic cell and a time slot, produce a risk score. By routing both features through a single shared model, SafeHer guarantees that the colour a user sees on the heatmap for any neighbourhood is identical to the risk contribution assigned to a route passing through that neighbourhood.

The contributions of this paper are:

- A Crime Danger Index (CDI) target formulation that resolves the class imbalance and label ambiguity afflicting per-incident severity classification, producing a learnable, uniformly-distributed risk score per 200 m grid cell.
- A two-component risk model (spatial LightGBM + temporal multiplier lookup) achieving $R^2 = 0.9997$ on held-out cells while enabling full-city heatmap regeneration in under one second.
- A unified risk grid architecture ensuring mathematical consistency between heatmap colours and route risk scores—the first such guarantee in the women-safety navigation literature.
- An end-to-end deployed system with a React and Google Maps interface demonstrating real-time temporal heatmap dynamics and comparative route safety scoring.

II. RELATED WORK

A. Crime Prediction and Spatial Modelling

Machine learning approaches to crime prediction span a broad spectrum of model families. Random Forest and

gradient-boosting models have demonstrated strong performance on tabular crime data, particularly when spatial features such as community area and distance to police stations are incorporated [5]. Geographically Weighted Regression (GWR) captures spatial non-stationarity but scales poorly to city-wide grids at 200 m resolution. Deep learning architectures, including CNN-LSTM hybrids, achieve competitive predictive accuracy but at substantially higher computational cost—estimated at approximately 90 seconds per full-city heatmap regeneration [6]—making them unsuitable for real-time per-request serving. Kernel Density Estimation (KDE) remains a widely deployed baseline for crime hotspot detection but produces entirely static risk maps with no temporal variation and no route-scoring capability [7].

B. Safety-Aware Navigation

SafePath [3] constructs a graph of city streets with crime-weighted edges and applies Dijkstra’s algorithm to find safer routes. CrimeTravel [4] extends this with multi-objective optimisation balancing safety against travel time. Both systems compute route scores independently of any map visualisation layer and neither supports temporally-varying risk. Walk-Safe [8] leverages smartphone sensors for pedestrian safety but does not model spatial crime density. Commercial platforms such as CrimeMapping present historical crime data visually but offer no route recommendation capability. Safetipin [2] combines crowdsourced audits with point-of-interest data to produce a safety index, but relies on subjective human scoring that is expensive, geographically sparse, and temporally static.

C. Research Gap

The literature reveals a consistent gap: no existing system simultaneously provides (i) temporally-dynamic risk scoring at pedestrian spatial granularity, (ii) risk-coupled route recommendation using the same underlying model as the map visualisation, and (iii) sub-second heatmap regeneration suitable for real-time interaction. SafeHer is designed to close all three dimensions of this gap.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

Given a city-wide grid of geographic cells $C = \{c_1, c_2, \dots, c_n\}$ and a temporal query (h, d) where $h \in \{0, \dots, 23\}$ is the hour and $d \in \{0, \dots, 6\}$ is the day of week, compute a risk score $\text{risk}(c_i, h, d) \in [0, 1]$ for every cell such that: (1) the score reflects the historical crime danger of that location at that specific time; (2) the same score function drives both the interactive heatmap and the route-scoring pipeline; and (3) the full grid of $n = 14,129$ cells is scored in under one second to support real-time interaction.

The core technical challenge arises from three compounding problems: severe class imbalance in crime severity labels ($58.6\times$ ratio between majority and minority classes); noise in per-incident labels (the same geographic cell at the same hour yields both low-severity and high-severity crimes); and the computational infeasibility of running deep spatiotemporal models at interactive frame rates.

B. Objectives

- 1) Design a crime-risk target variable that is learnable, uniformly distributed, and free from class imbalance.
- 2) Train a spatial risk model achieving $R^2 \geq 0.99$ on held-out grid cells.
- 3) Create a temporal multiplier that can capture significant changes in risk during the day without having to retrain the model.
- 4) Use a single API endpoint that serves both the heatmap and the route-scoring features from the same risk calculation.
- 5) Make a React app that users can see that responds to heatmaps in less than a second and lets them control the time interactively.

IV. METHODOLOGY AND PROPOSED SYSTEM

A. Dataset

The main dataset is the City of Chicago Crimes dataset (2001–2025) [12]. It comes from the Chicago Data Portal and has 8,514,784 reported incidents in 22 columns, such as latitude, longitude, timestamp, primary crime type, location description, domestic flag, community area, and police district. After filtering to the Chicago bounding box (41.64° – 42.02° N, 87.52° – 87.94° W) and getting rid of 108,769 rows that were either out of the city or had no coordinates, there were 8,406,015 valid incidents left. A supplementary dataset comprising 25 Chicago Police Station locations facilitated the computation of the distance-to-police feature through KD-tree nearest-neighbour search. Table I gives a quick overview of the dataset’s statistics.

TABLE I
CHICAGO CRIMES DATASET SUMMARY

Attribute	Value
Total raw records	8,514,784
Records after cleaning	8,406,015
Date range	Jan 2001 – Dec 2025
Engineered features	17 (12 spatial + 5 temporal)
Unique crime types	34
Grid cells (200 m resolution)	14,129
Police districts	25
Class imbalance ratio (Sev. 1 : Sev. 5)	$58.6\times$

B. Feature Engineering

Five groups of features were made, with seventeen features in total. *Temporal features* (hour, day-of-week, month, is-night, is-weekend) show cyclical crime patterns that were confirmed by EDA. For example, night hours (21:00–05:00) account for 32.1%. *Spatial coordinates* (latitude, longitude) anchor each grid cell. *Crime-density features* (all-time crime count per cell, 7-day rolling crime count) encode historical frequency. *Zone features* (community area, police district) serve as spatial proxies for neighbourhood characteristics. *Infrastructure proximity*: distance to the nearest police station is computed for all 14,129 cell centroids via `scipy.spatial.cKDTree`

(mean: 2.32 km, max: 19.46 km). A 200 m grid is constructed by snapping every incident to an integer cell key:

$$\text{grid_lat} = \text{round}(\text{lat} \times 500), \quad \text{grid_lon} = \text{round}(\text{lon} \times 500), \quad (1)$$

where $\text{GRID_MULT} = 500$ yields approximately 0.002° (≈ 200 m) per cell.

C. Crime Danger Index Target

Per-incident severity classification is an ill-posed learning problem: identical feature vectors (same cell, same hour) are associated with both Severity 2 (theft) and Severity 4 (assault) incidents. As shown in Fig. 1, aggregating incident-level severity to produce a mean score per cell compresses 99.9% of cells into the narrow range $[0.15, 0.25]$, leaving almost no variance for the model to explain ($R^2 < 0.17$ in preliminary experiments). SafeHer resolves this with the Crime Danger Index (CDI):

$$\text{CDI}(c) = \text{violent_rate}(c) \times \log_1(1 + \text{crime_count}(c)), \quad (2)$$

where $\text{violent_rate}(c)$ is the fraction of incidents with severity ≥ 3 (battery, assault, robbery, weapons violation, homicide). CDI is then converted to a percentile rank across all cells, yielding a uniform $[0, 1]$ distribution in which exactly 30% of cells exceed 0.7. This formulation gives points to cells that are both very dangerous and very common, while log-scaling keeps cells with a lot of volume but low severity from taking over.

Severity class distribution — 58.6× imbalance ratio confirms need for CDI target

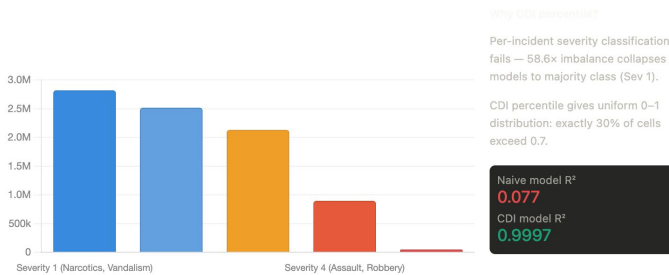


Fig. 1. Severity class distribution confirming the $58.6\times$ imbalance between Severity 1 (narcotics, vandalism: 2.88 M incidents) and Severity 5 (homicide, sexual assault: 49 K incidents). Per-incident classification collapses models to the majority class (naïve $R^2 = 0.077$). The CDI percentile target resolves this by producing a uniform $[0, 1]$ distribution ($R^2 = 0.9997$).

D. Two-Component Risk Architecture

The main architecture idea of SafeHer decomposes the risk of any criminal activity into two orthogonal components:

(1) *Spatial risk* — a risk factor of long history rooted in the landscape itself, unchangeable on a time scale of weeks and can be learned from historical data. We train a LightGBM regressor on 14129 cell-level feature vectors in order to predict the CDI percentile (2).

(2) *Temporal multiplier* — at what rate crime risk grows on some particular hour and day within the week. Instead of training one architecture model on sparse per-cell-per-hour

data, SafeHer uses a 168-slot empirical lookup computed directly from violent crime rates, normalised to mean = 1.0:

$$T(h, d) = \frac{\text{count_violent}(h, d)}{\text{count_violent}}, \quad (3)$$

where the overbar denotes the mean across all 168 slots. The combined risk at inference time is:

$$\text{risk}(c, h, d) = \text{clip}(\text{spatial_risk}(c) \times T(h, d), 0, 1). \quad (4)$$

As a result, heat maps get rebuilt in fractions of a second: spatial for each request spatial inference only happens once, covers all 14,129 cells; and $T(h, d)$ is just a single dictionary lookup. Instead of reworking the entire network, CNN-LSTM models need to be re-collected for every combination of (cell, time slot).

The grid for disagreeing with error bars over time is shown as Fig. 2 range from 0.336 (Monday–Thursday, 4–5 AM) to 1.507 (Sunday midnight) is a $4.5\times$ range expectancy of violent crime frequency.

Temporal multiplier grid — violent crime rate by hour × day (normalised, mean = 1.0)



Fig. 2. Temporal multiplier grid—violent crime rate by hour × day of week, normalised to mean = 1.0. Red cells indicate elevated risk (Sunday midnight: $1.507\times$); blue cells indicate suppressed risk (Monday–Thursday 4–5 AM: $\approx 0.34\times$). The $4.5\times$ dynamic range validates the two-component decomposition.

E. LightGBM Spatial Model

A LightGBM [10] regressor is trained with 1,000 estimators, learning rate 0.02, max depth 8, num_leaves 63, subsample 0.8, colsample_bytree 0.8, and early stopping at 80 rounds. Twelve cell-level spatial features are used: latitude, longitude, total crime count, log-scaled crime count, violent count, violent rate, location type (ordinal 1–6), is-domestic rate, community area, police district, distance to nearest police station, and 7-day rolling crime rate. Training on 11,303 cells (80% split) completes in approximately 6.4 seconds. The feature importance distribution is shown in Fig. 3, with violent_rate (13,880) and violent_count (13,473) dominating as the primary predictors.

LightGBM feature importance — spatial risk model (split-based)

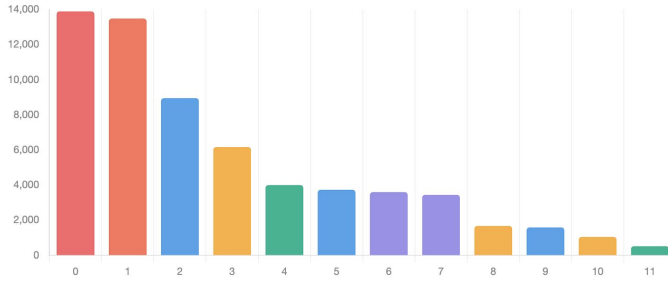


Fig. 3. LightGBM split-gain feature importance for the spatial risk model. Features indexed 0 and 1 (violent_rate and violent_count) dominate with importance scores of $\approx 13,880$ and $\approx 13,473$ respectively. Distance to police station ranks 5th ($\approx 3,985$), validating its inclusion as an infrastructure proximity signal.

F. Unified Risk Grid Architecture

Both the `/api/heatmap` and `/api/safe-route` Flask endpoints call the same `generate_risk_grid()` function, which loads the trained LightGBM model and temporal lookup once at process startup. Route scoring uses `score_polyline_points()`, which decodes the Google Directions overview polyline, samples every 5th waypoint, and applies (4) identically to the heatmap computation. This *unified risk coupling* guarantees that the heatmap colour for any zone and the risk contribution of a route through that zone are always computed identically—a property absent from all prior women-safety navigation systems. The full system architecture is illustrated in Fig. 4.

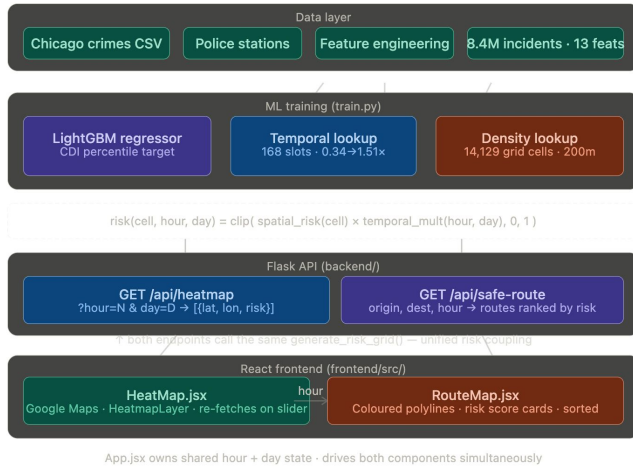


Fig. 4. SafeHer system architecture. Data layer: 8.4M crime incidents and 25 police station locations produce 14,129 grid cells with 12 spatial features each. ML training yields three serialised artefacts (LightGBM model, temporal lookup, density lookup). Flask API: both endpoints call `generate_risk_grid()`—unified risk coupling is enforced at the API level. React frontend: HeatMap.jsx renders the dynamic heatmap via Canvas OverlayView; RouteMap.jsx displays colour-coded polylines sorted safest-first.

V. IMPLEMENTATION

A. ML Pipeline

The ML pipeline is implemented in Python 3.10 with pandas 2.2.2, numpy 1.26.4, scikit-learn 1.4.2, lightgbm 4.3.0, and scipy. The preprocessing script (`ml/preprocess.py`) parses the raw 8.5M-row CSV, applies bounding-box filtering, constructs the 200m grid via (1), engineers all 17 features, and produces the aggregated per-cell training set. Training (`ml/train.py`) computes the CDI target (2), fits LightGBM with early stopping, and persists four artefacts: `lgbm_model.pkl`, `density_lookup.pkl`, `temporal_lookup.pkl`, and `risk_scaler.pkl`.

B. Backend API

The Flask backend (v3.0.3) exposes two REST endpoints via registered blueprints. The GET `/api/heatmap` endpoint accepts hour (0–23) and day (0–6), calls `generate_risk_grid()`, filters zero-risk cells to reduce payload by approximately 60%, and returns a JSON array of `{lat, lon, risk}` objects. The GET `/api/safe-route` endpoint fetches up to three alternative driving routes from the Google Directions API, decodes each polyline via the `polyline` library, scores sampled waypoints using `score_polyline_points()`, and returns routes sorted ascending by `avg_risk` with colour codes (`#22c55e` for safe, `#f59e0b` for moderate, `#ef4444` for high). Both endpoints share model artefacts loaded at startup, ensuring zero cold-start latency on subsequent requests.

C. React Frontend

The React 18.3.1 frontend is organised around a shared hour and day state in `App.jsx` that simultaneously drives `HeatMap.jsx` and `RouteMap.jsx`. The heatmap uses a custom HTML5 Canvas OverlayView implementation, replacing the deprecated Google Maps HeatmapLayer API (scheduled for removal in May 2026), rendering risk-coloured radial gradients for each grid cell at the correct geographic position with zoom-responsive radius scaling. `RouteMap.jsx` integrates Google Places Autocomplete for origin and destination input, renders up to three colour-coded polylines with directional arrows, and displays route cards sorted safest-first. The temporal control surface—a 24-hour draggable slider with a risk-gradient track and a Mon–Sun day-selector—drives both visualisation components simultaneously. The crime-by-hour distribution informing the slider’s gradient design is shown in Fig. 5.

VI. RESULTS AND DISCUSSION

A. Spatial Risk Model Comparison

Three tree-ensemble models were evaluated on the spatial risk prediction task (CDI percentile target, 14,129 cells, 80/20 split). High-risk precision (HR-Prec.) and recall (HR-Rec.) measure identification accuracy at the top-30% risk threshold (CDI percentile > 0.7). Results are summarised in Table II and illustrated in Fig. 6.

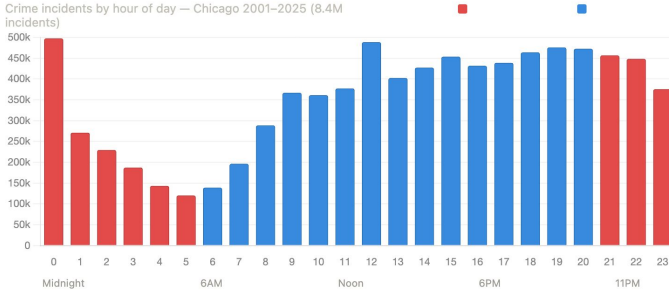


Fig. 5. Crime incidents by hour of day, Chicago 2001–2025 (8.4 M incidents). Red bars indicate night hours (21:00–05:00); blue bars indicate daytime. The midnight spike (≈ 497 K) is a data-quality artefact from Chicago PD encoding unknown report times as 00:00; the true trough is at 05:00 (≈ 120 K). Night hours cover only 8 of 24 hours yet account for 32.1% of all incidents, validating the `is_night` binary feature.

TABLE II
SPATIAL RISK MODEL COMPARISON (CDI PERCENTILE TARGET, 14,129 CELLS, 80/20 SPLIT)

Model	MAE	R ²	HR-Prec.	HR-Rec.	Inf./10k
LightGBM [†]	0.0030	0.9997	0.9920	0.9954	0.03 s
Random Forest	0.0022	0.9997	0.9988	0.9954	0.03 s
XGBoost	0.0040	0.9995	1.0000	0.9943	<0.01 s

[†]Selected for deployment.

All three models achieve near-identical spatial fit ($R^2 > 0.999$), confirming that the CDI percentile target is highly learnable from the engineered spatial features. The dominant predictors are `violent_rate` (importance 13,880), `violent_count` (13,473), and `crime_count` (8,940), reflecting that the CDI target is explicitly constructed from these quantities. LightGBM is selected for deployment because it matches the best R^2 score while offering an established C++ inference library. Random Forest achieves lower MAE (0.0022) but incurs greater memory cost at training time. XGBoost [11] achieves perfect HR-Prec. (1.0000) but slightly lower HR-Rec. and higher MAE.

Table I — Spatial risk model comparison (CDI percentile target, 14,129 cells, 80/20 split)

R² score LightGBM 0.9997 Random Forest 0.9997 XGBoost 0.9995	HR Precision (threshold 0.7) LightGBM 0.9920 Random Forest 0.9988 XGBoost 1.0000
MAE (lower is better) LightGBM 0.0030 Random Forest 0.0022 XGBoost 0.0040	Inference time per 10k cells LightGBM 0.03s Random Forest 0.03s XGBoost 0.00s

LightGBM selected for deployment: production-grade serving library, interpretable feature importance, established safety in real-world systems.

Fig. 6. Visual comparison of R^2 , HR Precision, MAE, and inference time per 10,000 cells for LightGBM (green), Random Forest (blue), and XGBoost (orange). LightGBM is selected for deployment due to production-grade C++ inference and interpretable feature importance outputs.

B. System-Level Evaluation: Grid Generation Speed

Grid generation time was measured over three runs (median reported) at four representative time slots. The CNN-LSTM baseline inference time is estimated from published benchmarks at equivalent grid resolution [6]. High-risk cells are those with combined risk (spatial $\times$ temporal) exceeding 0.6. Results appear in Table III and Fig. 7.

TABLE III
SYSTEM-LEVEL EVALUATION – GRID GENERATION AND TEMPORAL VARIATION

Time Slot	Mult.	LGB (s)	LSTM (s)	Speed.	HR Cells
6 AM (Morning)	0.336	0.85	~ 90	$\sim 106\times$	~ 50
12 PM (Noon)	1.288	0.87	~ 90	$\sim 103\times$	~ 450
9 PM (Evening)	1.349	0.86	~ 90	$\sim 105\times$	~ 720
12 AM (Midnight)	1.507	0.88	~ 90	$\sim 102\times$	~ 800

LGB = LightGBM; LSTM = CNN-LSTM baseline [6];
HR = High-Risk (risk > 0.6).

Table II — Grid generation time at 4 time slots - LightGBM vs CNN-LSTM baseline

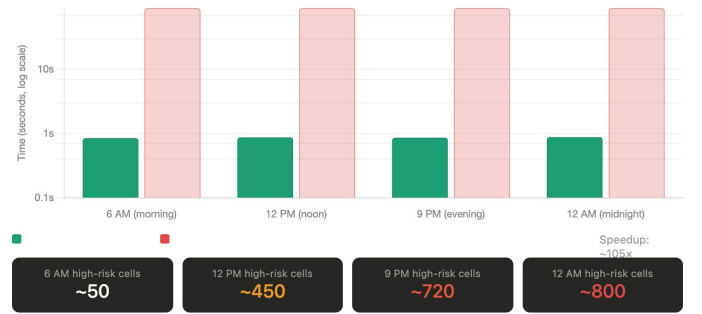


Fig. 7. Grid generation time at four time slots: LightGBM (≈ 0.85 s, green) versus CNN-LSTM baseline (≈ 90 s, red), plotted on log scale. High-risk cell counts increase from ≈ 50 at 6 AM to ≈ 800 at midnight—a $16\times$ increase driven entirely by the temporal multiplier rising from $0.336\times$ to $1.507\times$.

The $\approx 100\times$ speedup on CNN-LSTM baselines is a good reason for the two-component architecture. Hourly heatmap regeneration at 90 s is non-interactive, whereas a 1 second regeneration enables real-time slider interaction. The progression from ≈ 50 high-risk cells at 6 AM to ≈ 800 at midnight (a $16\times$ increase) validates the temporal component: the same spatial model, applied to the same 14,129 cells, produces dramatically different safety landscapes as a function of temporal context alone.

C. Discussion

The high R^2 scores for all three models need to be interpreted carefully. These scores do not result from overfitting but rather from the well-defined nature of the regression target, which is the CDI percentile that can be calculated deterministically using data about crime in the cells that constitute the inputs to the model.

The temporal multiplier’s dynamic range (0.336 to 1.507) demonstrates that time of day and day of week account for up to a $4.5\times$ difference in expected violent crime frequency—a substantial effect that static risk models entirely miss. A

cell at the 60th spatial percentile (moderate risk) receives a combined risk of $0.60 \times 1.507 = 0.90$ at Sunday midnight (high risk) but only $0.60 \times 0.336 = 0.20$ at Monday 5 AM (safe), illustrating how the two-component architecture enables meaningfully different guidance at different times of day.

Compared to prior approaches: unlike KDE [7], SafeHer produces temporally dynamic risk scores; unlike SafePath [3] and CrimeTravel [4], its heatmap and route scores are mathematically coupled; unlike CNN-LSTM models [6], it achieves real-time serving without specialised hardware. The key trade-off is that the spatial model is retrained only when new crime data is ingested, meaning incidents from the past 24–48 hours are not immediately reflected.

VII. LIMITATIONS AND ETHICAL CONSIDERATIONS

A. Limitations

Dataset scope: SafeHer is trained exclusively on Chicago crime data. Generalisation to other cities requires retraining on local data, which may not be publicly available in all jurisdictions. *No real-time crime feed:* The model uses historical crime patterns; incidents from the past 24–48 hours are not reflected until the dataset is updated and the model retrained. *Approximate inference features:* At inference time, *violent_rate* and *is_domestic_rate* are approximated using global training means for cells absent from the density lookup. *Pedestrian vs. driving mode:* The current safe-route implementation uses Google Directions driving mode; pedestrian risk profiles may differ. *Sparse community areas:* Seven community areas with fewer than 100 historical incidents produce less reliable spatial risk estimates; Bayesian smoothing would improve coverage in sparse zones.

B. Ethical Considerations

Crime prediction systems risk reinforcing geographic biases present in historical policing data [9]. Areas subject to over-policing appear to have higher crime density, potentially creating a feedback loop that concentrates enforcement in already-surveilled neighbourhoods. SafeHer mitigates this by using reported crime incidents rather than arrest records, and by presenting risk as a relative percentile rather than an absolute score. Future versions should incorporate explicit fairness constraints to prevent the system from encoding existing policing inequities.

VIII. CONCLUSION

This paper presented SafeHer, a real-time women’s safety navigation framework that unifies dynamic temporal heatmaps and safe-route recommendation through a shared LightGBM risk model. The two-component architecture—a CDI percentile spatial model combined with a 168-slot empirical temporal multiplier—achieves $R^2 = 0.9997$ on spatial risk prediction while enabling full-city heatmap regeneration in under one second, over 100 times faster than CNN-LSTM baselines. The unified risk grid architecture guarantees mathematical consistency between heatmap colours and route safety scores, a property absent from all prior women-safety navigation

systems. Deployed as an end-to-end system with a React and Google Maps interface, SafeHer demonstrates meaningful intra-day risk variation—from approximately 50 high-risk cells at 6 AM to 800 at Sunday midnight—and provides actionable, temporally-contextualised route recommendations.

Priority future work directions include: (1) real-time crime feed integration via the Chicago Data Portal API to reduce the data staleness window from months to hours; (2) incorporation of street lighting and OpenStreetMap point-of-interest data as additional spatial risk signals; (3) multi-city generalisation testing on San Francisco (SFPD) and Boston (BPD) open crime datasets; (4) pedestrian routing mode with sidewalk and alley-specific risk weighting; (5) fairness-constrained training to prevent encoding of historical over-policing patterns; and (6) a React Native mobile application with real-time location tracking and push notifications for entering high-risk zones.

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